

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 11

Level 1 — Beginner · Fractions II

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 1 — Beginner	Topic Focus	Fractions II — Equivalent Fractions & Comparing
Questions	20	Time Allowed	30 minutes
Total Marks	20	Worksheet No.	11 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 11 deepens fraction sense. The child builds and recognises equivalent fractions, simplifies to lowest form, compares fractions with the same numerator or denominator, and adds and subtracts like fractions — including 'how much makes a whole?' questions, a favourite Olympiad pattern. Real-life contexts (chocolate, bottles, ropes) keep it concrete.

Learning Objectives

- Build equivalent fractions by multiplying top and bottom by the same number.
- Simplify fractions to their lowest form.
- Compare and order like fractions, and unit fractions with different denominators.
- Add and subtract fractions with the same denominator.
- Find the fraction needed to complete one whole.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

Q1. Which fraction is **equivalent** (equal) to $\frac{1}{2}$?

- (A) $\frac{2}{3}$ (B) $\frac{3}{6}$
(C) $\frac{1}{4}$ (D) $\frac{2}{6}$

Q2. Which fraction is equivalent to $\frac{2}{3}$?

- (A) $\frac{4}{6}$ (B) $\frac{3}{4}$
(C) $\frac{2}{6}$ (D) $\frac{4}{9}$

Q3. What is $\frac{1}{5} + \frac{2}{5}$?

- (A) $\frac{3}{10}$ (B) $\frac{3}{5}$
(C) $\frac{2}{25}$ (D) $\frac{1}{5}$

Q4. What is $\frac{5}{8} - \frac{2}{8}$?

- (A) $\frac{3}{8}$ (B) $\frac{7}{8}$
(C) $\frac{3}{16}$ (D) $\frac{2}{8}$

Q5. Which is GREATER: $\frac{3}{7}$ or $\frac{5}{7}$?

- (A) $\frac{3}{7}$ (B) $\frac{5}{7}$
(C) Both equal (D) Cannot compare

Q6. Fill the blank: $\frac{1}{4} = \frac{\quad}{8}$

- (A) 1 (B) 2
(C) 3 (D) 4

Q7. What is $\frac{2}{9} + \frac{4}{9} + \frac{3}{9}$?

- (A) $\frac{9}{27}$ (B) 1
(C) $\frac{2}{9}$ (D) $\frac{8}{9}$

Q8. Simplify $\frac{4}{8}$ to its simplest form.

- (A) $\frac{2}{4}$ (B) $\frac{1}{2}$
(C) $\frac{4}{8}$ (D) $\frac{1}{4}$

Q9. Riya ate $\frac{2}{6}$ of a chocolate and Diya ate $\frac{3}{6}$ of the SAME chocolate. Who ate MORE?

- (A) Riya (B) Diya
(C) Both equal (D) Cannot say

Q10. What fraction must be ADDED to $\frac{3}{10}$ to make **1 whole**?

- (A) $\frac{3}{10}$ (B) $\frac{7}{10}$
(C) $\frac{10}{10}$ (D) $\frac{1}{10}$

Q11. Which fraction is GREATER: $\frac{1}{3}$ or $\frac{1}{6}$?

- (A) $\frac{1}{3}$ (B) $\frac{1}{6}$
(C) Both equal (D) Cannot say

Q12. Which of these is in its SIMPLEST form?

- (A) $\frac{2}{4}$ (B) $\frac{3}{9}$
 (C) $\frac{3}{5}$ (D) $\frac{4}{6}$

Q13. A water bottle is $\frac{3}{4}$ full. Aman drinks $\frac{1}{4}$ of the bottle. How full is the bottle now?

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$
 (C) $\frac{1}{4}$ (D) $\frac{4}{4}$

Q14. Which pair shows EQUIVALENT fractions?

- (A) $\frac{1}{2}$ and $\frac{2}{5}$ (B) $\frac{2}{3}$ and $\frac{6}{9}$
 (C) $\frac{1}{4}$ and $\frac{2}{6}$ (D) $\frac{3}{4}$ and $\frac{4}{3}$

Q15. What is $\frac{7}{12} - \frac{5}{12}$?

- (A) $\frac{2}{12}$ (B) $\frac{12}{12}$
 (C) $\frac{2}{24}$ (D) $\frac{12}{2}$

Q16. Arrange from GREATEST to SMALLEST: $\frac{4}{9}$, $\frac{7}{9}$, $\frac{2}{9}$

- (A) $\frac{4}{9}$, $\frac{7}{9}$, $\frac{2}{9}$ (B) $\frac{7}{9}$, $\frac{4}{9}$, $\frac{2}{9}$
 (C) $\frac{2}{9}$, $\frac{4}{9}$, $\frac{7}{9}$ (D) $\frac{7}{9}$, $\frac{2}{9}$, $\frac{4}{9}$

Q17. Aarav coloured $\frac{1}{2}$ of a page. His sister coloured $\frac{2}{4}$ of an identical page. Who coloured MORE?

- (A) Aarav (B) His sister
 (C) Both coloured the same (D) Cannot say

Q18. What is $1 - \frac{3}{5}$ (one whole minus three-fifths)?

- (A) $\frac{3}{5}$ (B) $\frac{2}{5}$
 (C) $\frac{1}{5}$ (D) $\frac{4}{5}$

Q19. Fill the blank: $\frac{3}{4} = \frac{9}{\underline{\hspace{1cm}}}$

- (A) 8 (B) 10
 (C) 12 (D) 16

Q20. A rope is divided into 10 equal parts. 4 parts are painted red, 3 parts blue, and the rest green. What fraction is GREEN?

- (A) $\frac{3}{10}$ (B) $\frac{4}{10}$
 (C) $\frac{7}{10}$ (D) $\frac{10}{10}$


ANSWER KEY & DETAILED SOLUTIONS
Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	A	B	A	B	B	B	B	B	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	C	A	B	A	B	C	B	C	A

Step-by-Step Solutions
Q1. Correct answer: (B)

Multiply top and bottom of $\frac{1}{2}$ by 3: $\frac{1 \times 3}{2 \times 3} = \frac{3}{6}$. Equivalent fractions show the same amount.

Q2. Correct answer: (A)

Multiply top and bottom by 2: $\frac{2 \times 2}{3 \times 2} = \frac{4}{6}$.

Q3. Correct answer: (B)

Same denominators: add only the numerators. $1 + 2 = 3$, so the answer is $\frac{3}{5}$.

Q4. Correct answer: (A)

Same denominators: subtract the numerators. $5 - 2 = 3 \rightarrow \frac{3}{8}$.

Q5. Correct answer: (B)

Same denominator $\rightarrow$ compare numerators: $5 > 3$, so $\frac{5}{7}$ is greater.

Q6. Correct answer: (B)

Multiply top and bottom of $\frac{1}{4}$ by 2: $\frac{1}{4} = \frac{2}{8}$.

Q7. Correct answer: (B)

$2 + 4 + 3 = 9$, so the sum is $\frac{9}{9}$, which is **1 whole**.

Q8. Correct answer: (B)

Divide top and bottom by 4: $\frac{4 \div 4}{8 \div 4} = \frac{1}{2}$. ($\frac{2}{4}$ is equal but not the simplest form.)

Q9. Correct answer: (B)

Same denominator: $\frac{3}{6} > \frac{2}{6}$, so **Diya** ate more.

Q10. Correct answer: (B)

A whole is $\frac{10}{10}$. $\frac{10}{10} - \frac{3}{10} = \frac{7}{10}$.

Q11. Correct answer: (A)

Same numerator $\rightarrow$ smaller denominator is greater: thirds are bigger pieces than sixths, so **$\frac{1}{3}$** .

Q12. Correct answer: (C)

$\frac{3}{5}$ cannot be divided down further (3 and 5 share no common factor except 1). The others simplify: $\frac{2}{4} \rightarrow \frac{1}{2}$, $\frac{3}{9} \rightarrow \frac{1}{3}$, $\frac{4}{6} \rightarrow \frac{2}{3}$.

Q13. Correct answer: (A)

$\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$. The bottle is now half full.

Q14. Correct answer: (B)

$\frac{2}{3} \times 3$ top and bottom $\rightarrow \frac{6}{9}$, so **$\frac{2}{3}$ and $\frac{6}{9}$** are equivalent. ($\frac{3}{4}$ and $\frac{4}{3}$ are reciprocals, not equal.)

Q15. Correct answer: (A)

Same denominators: $7 - 5 = 2 \rightarrow \frac{2}{12}$ (which simplifies to $\frac{1}{6}$).



Q16. Correct answer: (B)

Same denominator → order by numerators: $7/9 > 4/9 > 2/9$.

Q17. Correct answer: (C)

$2/4$ simplifies to $1/2$, so both coloured exactly **the same** amount.

Q18. Correct answer: (B)

One whole = $5/5$. $5/5 - 3/5 = 2/5$.

Q19. Correct answer: (C)

The numerator went $3 \rightarrow 9$ (multiplied by 3), so the denominator must also be $\times 3$: $4 \times 3 = 12$.

Q20. Correct answer: (A)

Red + blue = $4 + 3 = 7$ parts. Green = $10 - 7 = 3$ parts → **3/10**.

